Practical Radiation Oncology (2012) 2, 157–163

practical radiation oncology
pro

www.practicalradonc.org

Original Report

A 2-year review of recent Nuclear Regulatory Commission events: What errors occur in the modern brachytherapy era?

Susan Richardson PhD*

Department of Radiation Oncology, Mallinckrodt Institute of Radiology, Washington University School of Medicine, St Louis, Missouri

Received 1 July 2011; revised 10 August 2011; accepted 23 August 2011

Abstract

Purpose: To perform a retrospective analysis of recently reported brachytherapy errors to the Nuclear Regulatory Commission and to compare with historical trends.

Methods: All events reported in the 2-year period from January 1, 2009 to December 31, 2010 were categorized and analyzed. The 4 main areas of dose delivery were Gamma Knife radiosurgery, therapeutic radiopharmaceutical administration, high-dose-rate brachytherapy, and low-dose-rate brachytherapy. The different types of errors were wrong site, wrong dose, unintended exposure, lost or leaking source, or other. The causes of events were specified as the following: communication errors, equipment malfunction, human error, lack of training, or miscellaneous.

Results: One hundred and forty-seven events were found in the 2-year period. This error reporting rate far surpasses previous reports. The greatest number of events reported was for low-dose-rate brachytherapy, and the most common cause of error was human error. Wrong dose was the error that occurred most often, followed by wrong site.

Conclusions: Very simple treatment errors, such as wrong patient, or wrong side of patient treated, are still occurring. Newer, complex deliveries such as high-dose-rate partial breast irradiation and low-dose-rate prostate brachytherapy also had a large number of events reported in this sampling. This report can help institutions establish needs for quality assessment and quality control processes.

Introduction

Patient safety and error reduction should be an important goal in any medical facility. In radiation oncology, complexities such as advanced equipment mechanics, large treatment dose, or complicated patient setup should be carefully scrutinized for errors. Gamma Knife (Elekta, Stockholm, Sweden) stereotactic radiosurgery, brachytherapy, and therapeutic radiopharmaceutical administration are often more highly regulated and monitored

Parts of this research were presented in oral form at the 2011 annual meeting of the American Brachytherapy Society, April 14-16, San Diego, CA.

Conflicts of interest: None.

* Corresponding author. Department of Radiation Oncology, Campus Box 8224, Washington University School of Medicine, St Louis, MO 63110.

E-mail address: srichardson@radonc.wustl.edu (S. Richardson).

1879-8500/$ – see front matter © 2012 American Society for Radiation Oncology. Published by Elsevier Inc. All rights reserved.
doi:10.1016/j.prro.2011.08.004

than linear accelerator-based therapies due to requirements by the Nuclear Regulatory Commission (NRC).

Brachytherapy errors made the *New York Times* front page in 2009.[1] Since then, there has been a surge of publications, meeting sessions, and informal discussions among radiotherapy practitioners about error reduction and prevention. One methodology that has been employed for decades in other fields is failure mode and effects analysis (FMEA). The American Association of Physicists in Medicine is currently producing a report which explicitly describes the application of FMEA to radiotherapy.[2] While prevention of radiotherapy errors is ideal, another methodology is simply to review errors that have already occurred and determine what processes caused those errors through root cause analysis.

In 1994, Ostrom et al investigated brachytherapy events (then called misadministrations) and determined the following 8 causes of error: lack of policies and procedures; lack of radiation safety oversight; lack of training and experience; lack of supervision, errors in judgment, errors in interpretation, lack of communication, and hardware errors.[3] These causes are still prevalent today. In 2003, Thomadsen et al retrospectively investigated 134 brachytherapy events that occurred from 1980 to 2001.[4] That publication described several aspects of various brachytherapy errors, including number of reported events per year for high-dose-rate and low-dose-rate brachytherapy (HDBRT and LDBRT), specific causes for these events, and where in the treatment process these events occurred. However, the data for that publication are 10 to 30 years old and much has changed in the realm of brachytherapy treatment delivery. Most HDBRT treatments now rely on 3-dimensional imaging (computed tomography or magnetic resonance imaging) and computerized 3-dimensional treatment planning rather than orthogonal films and hand calculations for dwell times. Low-dose-rate prostate brachytherapy implant rates have increased significantly over the last decade, including the development of intraoperative dosimetry. Modern computers and technology are used in almost every aspect of treatment delivery. This poses the following questions: Does this technology reduce the brachytherapy error rate; and Are new sources of errors replacing old ones? This work will perform a retrospective review of brachytherapy events reported to the NRC in the modern brachytherapy era and seek to answer these questions.

Materials and methods

Source of data

All data in this analysis come from the online database of reported events to the NRC at www.nrc.gov.[5] The reported event dates reviewed were from January 1, 2009 to December 31, 2010. Similar to that described by Thomadsen et al,[4] the data provided in these reports suffer from both incompleteness and vagueness. Some items may also be retracted if the reporting institution believes a mistake was made in the reporting (ie, it was subsequently determined there was no reportable event). Items are often updated and appear in the database multiple times as more information becomes available. This is a side effect of the requirement that all errors be reported within 24 hours of discovery; it often takes several days for the sequence(s) of events relating to the error to be determined. In cases of multiple reports for an isolated incident, the event has only been counted as a single error. In cases of the event being retracted, the event will not be counted unless otherwise described. Some reported items may not actually be medical events in strict accordance with the definitions provided in 10 CFR 35.3045; however, these items may still be of interest. These items will be explicitly described separately and will be counted as a reported event. Events that were reported in the previous 2 years (2009 and 2010) may have occurred as early as 2003. With news about medical events making the front page of newspapers, many institutions have begun doing retrospective reviews of all their patient treatments. These retrospective reviews can result in medical errors being reported in the past 2 years even though the patient may have been treated many years prior.

Classification of errors

Four categories of dose delivery methods were tabulated, including HDRBT, LDRBT, Gamma Knife radiosurgery (GK), and radiopharmaceutical administration (RP). The errors tabulated only describe therapeutic use of radiation; ie, cases of errors reported while using radiopharmaceuticals for imaging purposes were not counted. The dose delivery methods for events that involve lost or missing sources or unintended exposures were counted under the main goal of the primary therapy used with the isotope. Only Co-60 or Gamma Knife radiosurgeries will be discussed as linac-based radiosurgery does not fall under the jurisdiction of the NRC.

Five different categories of error type were found, including wrong dose, wrong site, unintended exposure, lost or leaking sources, and miscellaneous. For each category, the main mechanism of failure was described if determinable. These included human error, lack of training, equipment malfunction, or communication issues. While this is not an exhaustive or an all-inclusive list of causes of medical events, it is a general category that can be used to perform FMEA. Additionally, most events probably have multiple causes and are a compilation of factors. If the NRC event report stated the cause of the problem (ie, the physicist did not have

Table 1 Summary of events reported from January 1, 2009 to December 31, 2010

Type of dose delivery (No. of events)	Type of error (No. of events)	Isotope (No. of events)
Gamma Knife (13)	Wrong site (7)	Co-60 (7)
	Wrong dose (5)	Co-60 (5)
	Unintended exposure (1)	Co-60 (1)
RP (34)	Wrong site (2)	I-125 (1)
		Y-90 (1)
	Wrong dose (22)	I-131 (9)
		Y-90 (13)
	Unintended exposure (8)	I-131 (8)
	Other (2)	P-32 (1)
		Y-90 (1)
HDR (33)	Wrong site (20)	Ir-192 (19)
		Y-90 (1)
	Wrong dose (9)	Ir-192 (5)
		Sr-90 (4)
	Unintended exposure (3)	Ir-192 (3)
	Other (1)	Ir-192 (1)
LDR (66)	Wrong site (6)	I-125 (3)
		Pd-103 (1)
		Cs-137 (2)
	Wrong dose (45)	I-125 (37)
		Pd-103 (4)
		Cs-131 (3)
		Variable (1)
	Lost or leaking source (13)	I-125 (9)
		Cs-137 (4)
	Unintended exposure (1)	Cs-137 (1)
	Other (1)	Cs-131 (1)
Unspecified (1)	Other (1)	Unspecified (1)

Left-hand column describes type of dose delivery employed when the error occurred. Middle column describes the nature of the event. Right-hand column shows the isotopes that were used during the event. Numbers in parentheses indicate the number of events reported. GK, Gamma Knife; HDR, high dose rate; LDR, low dose rate; RP, radiopharmaceutical.

training with Y-90 microspheres), the error was classified as such. Otherwise, the main cause was determined by the author. Lost sources are not classified as medical events; however, the processes involved are interesting and elucidative.

Additionally, a single tabulated event does not necessarily describe the treatment of a single patient. This is particularly true for the cases of LDRBT for prostate; some states (such as Ohio and Wisconsin) recently issued a mandatory review of all their previous brachytherapy prostate cases. Therefore, a single event could potentially include a dozen or more actual patient treatments. This is also the case for a systematic equipment failure such as a treatment planning software bug.

Results

Overview

In the time period from January 1, 2009 to December 31, 2010, 147 separate reported events were found. There were 13 GK events, 33 HDRBT events, 66 LDRBT, and 34 RP events. There was one additional event that was classified as unspecified because neither the isotope nor the procedure was described. The most commonly reported event was wrong dose, with 81 events being recorded. Wrong site occurred 35 times. Unintended exposures were counted 13 times, and a lost or leaking source had 14 occurrences. There were also 4 miscellaneous events found. A summary of event type, delivery method, and isotope can be found in Table 1. Wrong site occurred most frequently with HDRBT, and wrong dose occurred most often with LDRBT. Wrong dose was also the primary reported error with radiopharmaceutical delivery. Out of the GK events, wrong site and wrong dose were most frequent. Unintended exposures occurred most often with radiopharmaceutical delivery. For the case of LDRBT, the wrong dose classification could have easily been construed as wrong site; if the seeds were placed in the wrong area the target could have been under-dosed (wrong dose) and other regions near the prostate could be irradiated when they were not intended to be (wrong site). The error type was taken directly from the NRC's classification of the error.

Table 2 describes the primary causes of the reported events and the type of dose delivery that was affected.

Table 2 Causes of events (left) and type of dose delivery involved with the event (right)

Causes of event (No. of events)	Type of dose delivery (No. of events)
Communication issues (10)	LDR (3)
	RP (7)
Equipment malfunction (23)	GK(6)
	HDR (8)
	LDR (4)
	RPl (5)
Human error (97)	GK(7)
	HDR (20)
	LDR (58)
	RP (12)
Lack of training (12)	HDR (1)
	LDR (5)
	RP (6)
	Other (1)
Miscellaneous (5)	Other (5)

Numbers in parentheses indicate the number of events reported. GK, Gamma Knife; HDR, high dose rate; LDR, low dose rate; RP, radiopharmaceutical.

The dominant cause found for these events was human error with 97 events. Equipment malfunction occurred 23 times, followed by lack of training (12 events), and communication issues (10 events). Human error covers a wide range of subtypes (including blunders, rushing, confusion, inexperience, etc) and is generally thought of as the most difficult error to prevent. Any scenario that involved lost seeds was attributed to human error (blunder), but lack of training could easily have been chosen. Lack of training was selected only if an individual attempted to perform an action and did so incorrectly all the while thinking they did the action correctly. Equipment malfunctions were most common in GK and HDRBT deliveries and often included misuse of the head frame localization system and applicator use, respectively.

Category-specific events

A small sampling of events for each category will be described here.

Gamma Knife

One GK event resulted in an unintended exposure. This was initially caused by a couch sensor failing due to a software bug (according to the report). The emergency stops failed and the treating therapist had to enter the room and to manually close the GK doors to shield the source. Several instances of the head frame or fixation system malfunctioning were reported, including the head frame coming undone and causing the patient to be misaligned. There were 2 instances in which the incorrect side of the head was treated (right or left error), and one case where the incorrect cranial nerve was outlined on the magnetic resonance imaging and treated. The wrong site of a patient was treated when a physicist transposed the fixation coordinates of the localizer box and set the patient up to the coordinates (x,y,x) instead of (x,y,z).

Radiopharmaceutical: Thyroid brachytherapy

There were 8 cases of unintended exposure reported during this time period with radiopharmaceutical administrations. In 6 cases, the patient was receiving I-131 thyroid ablation, and the person unintentionally irradiated was the patient's fetus. In 4 of these cases, the patient was administered a pregnancy test and the patient tested negative directly before treatment. In 2 cases, the patient was already pregnant; one was approximately 6 months along when she received her therapy. These cases were assumed to be failures of communication because the patient either neglected or ignored the physician consult, which usually describes dangers to the unborn fetus with I-131 therapy. One patient who was already pregnant had received previous I-131 therapy and was not reconsulted prior to her subsequent therapy. Five cases involved the wrong dose of radiation being administered to an I-131 ablation patient due to human error. This occurred because of the following: (1) the vials of 2 patients were inadvertently switched; (2) the wrong activity of the isotope was ordered; or (3) the patient did not take all the capsules they were prescribed, resulting in an underdose.

Microsphere brachytherapy

For Y-90 treatments for liver cancer or metastases, a majority of the errors reported had to do with either equipment malfunction or misuse of the delivery equipment due to lack of training. There were 4 reported cases of leaking or occluded lines which would occur if the lines were not connected properly to the dose delivery unit. These all resulted in an underdose to the patient. Three errors were attributed to being unable to get the dose out of the vial; this can happen if the vial is not shaken properly before treatment delivery, or if the vial is not tapped to remove a clog. One error was reported in treatment of the incorrect lobe of the liver (right versus left).

High-dose-rate brachytherapy

There were 3 reports of unintended exposures resulting from the HDRBT source not retracting properly. One was during a source exchange, one was prior to a patient treatment during quality assurance, and one was during a patient treatment.

There were many events in which misuse of the applicator resulted in the wrong site or wrong dose to the patient. The most common error was either measuring the applicator length incorrectly or entering the length into the treatment planning system incorrectly. This was reported 11 times. The most common applicator in which the treatment length was incorrectly measured was with the use of partial breast irradiation applicators; there were 5 reported events that involved the catheter length being wrong and subsequent irradiation to the wrong part of the patient's breast. There were 2 instances where the distances were off by 10 cm, and 3 instances in the 2-3 cm range. In one case of a 10-cm positioning error, it was the first time the institution had used that particular applicator, and a representative of the vendor was present at the time of treatment. It was unclear how most of these particular errors were caught; however, one patient complained on her follow-up of redness to the breast. After retrospective review, her skin dose was determined to be greater than 68 Gy. Another site in which reporting incorrect treatment length was common was the liver; the duodenum is a winding structure and the dummy wire tends to stick, resulting in the wrong length of the applicator being measured. If imaging is not done to verify the location of the dummy wire, the wrong site of the patient will be treated.

Three errors were indicated due to the applicator being positioned incorrectly. In 2 of the treatments, the catheters or applicators moved during the treatment and were found to be in the incorrect location after the treatment was completed. In one case, X-rays were taken after the patient treatment which revealed the patient's tandem was 6-cm distal to the proper location.

There were several reports that dealt with software or treatment planning errors. One event which involved 5 patients was caught after a new consulting medical physics group did a retrospective review of all the HDRBT patients treated at the institution. They determined that the previous physicists had incorrectly interpreted the results of the treatment planning system. The original group had wrongly concluded that the treatment planning system would calculate the dwell time for all the prescribed fractions instead of a single fraction. They had divided the total time by the number of fractions (4 patients received 3 fractions, and one patient received 2 fractions) resulting in underdosages of 33% and 50%, respectively.

A wrong dose error was reported with Sr-90 brachytherapy to the heart. The typical treatments are either 23 Gy or 18.4 Gy and the physicist wrote down the incorrect time for the prescribed dose. Additionally, the written directive was not signed by the authorized user prior to the treatment.

Low-dose-rate brachytherapy

Five errors were reported with the use of Cs-137 for gynecologic brachytherapy. Two of these involved sources being lost for some time and eventually found wrapped up in the patient's dirty linens. Another 2 errors involved sources falling out but lying next to the patient and irradiating an unplanned area of the patient. A final error was attributed to a source that was leaking and contamination was found by a radiation survey in the treatment area, hallways, offices, and the supply rooms of the facility. Another reported event involving Cs-137 dealt with a missing storage safe that contained 14 sources. The safe had been moved to the basement of the facility during remodeling and upon completion of the construction the safe was nowhere to be found. An unintended exposure was reported after a visitor stayed overnight with a patient who had a Cs-137 implant on 2 separate occasions. The visitor was told he could stay 2 hours but did not comply with instructions.

Lost sources were common, especially when dealing with I-125 or Pd-103 sources for prostate implants. A few seeds would be lost during the sterilization process or in transit to or from sterilization. Additionally, "extra" seeds that were ordered for calibration purposes also seemed susceptible to being lost. One event reported an entire shielded Mick cartridge (Mick Radio-Nuclear Instruments, Inc, Mount Vernon, NY) of seeds was inadvertently thrown in the trash and was subsequently unable to be found. Seeds also went missing after a patient's implant.

The definition of wrong dose has caused much concern within the radiation oncology community lately. The NRC is currently using the definition of wrong (under) dose for LDRBT to be D90 < 80% (the dose to 90% of the target is less than 80% of the intended dose), and the definition of wrong site for LDRBT to be 20% of the seeds implanted outside of the prostate. With these definitions in mind, the distribution of reported D90 values is shown in Fig 1. Most D90s were greater than 70%, but the second most frequent event was 10% and below (that is, the dose encompassing 90% of the prostate was less than 10% of the prescribed dose). Not every event of wrong dose included the D90 value; these data are just for the incidents in which the value was reported. Additionally, in several events the D90 values for several patients at once were reported, hence the mismatch between numbers of reported events and frequency. Several institutions reported generic findings from their retrospective reviews, such as "10 patients had D90 < 80%." Other institutions had their own policies, such as D80 < 80%. One reported D80 = 55% and another reported V100 = 47% as a medical event. Three events indicated patients were "underdosed" but not by how much. Two events listed patients as being "overdosed" to the prostate by 21.4% and 21%, respectively. Others indicated medical events involving overdosing due to prescription error. Two events indicated an overdose of approximately 30% when the isotope activity was ordered in air-kerma strength (U) and the activity received was the same except in milliCuries. The mistake was not noticed until after the implant had occurred.

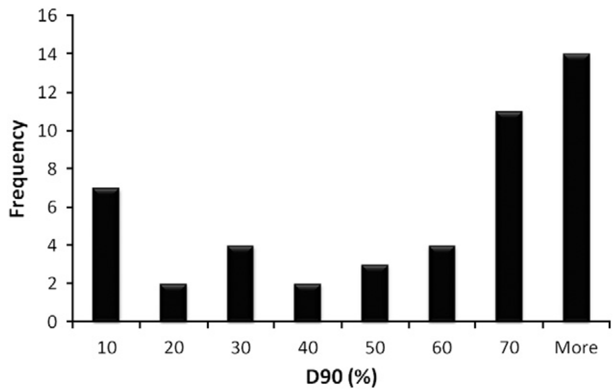

Figure 1 D90 values (dose %) for reported low-dose-rate (LDR) prostate brachytherapy events.

Discussion

One hundred and forty-seven events were found in the NRC database in a recent 2-year period. Thomadsen et al[4] found 134 events in a 21-year period (1980-2001) and Ostrom et al[3] found 35 in 1991 and 1992. This result could be thought of in 2 ways: (1) as a community we are committing many more brachytherapy errors (or treating many more patients with brachytherapy and maintaining the same error rate); or (2) error reporting is becoming more acceptable and more events are being reported. Either way, the number and severity of the reported events reveal a great need for addressing quality control and quality assurance processes. The event summaries provided here are provided for 2 purposes. One is to indicate which errors occur most often and why they occurred, to help with an FMEA. Second, each event provides a unique learning opportunity where new quality-assurance processes can be directly created and implemented in response to an event.

For example, for I-131 ablation, several patients became pregnant directly before or after their therapy. At our clinic, patients are consented and treated on the same day. While no specific information is available about policies of other institutions, it might be reasonable to assume that the female patients were asked if they could be pregnant. The patients answered no, human chorionic gonadotropin tests indicated they were not pregnant, and then their therapy was administered. It should be possible to consent and inform the patient a month ahead of time regarding the seriousness of their therapy and provide contraception if necessary. This could potentially reduce the number of unintended exposures with I-131 therapy.

For HDRBT, the most common error was incorrect applicator measurement. This is similar to the finding of Thomadsen et al. in which the most common step of failure was entering the treatment distance.[4] It could be reasoned that one way to reduce this error is to have multiple individuals measure an applicator treatment length and check its value in the treatment planning system. It would also be helpful if vendors would publicize the standard lengths of their applicators. For several partial breast applicator vendors, they explicitly refuse to publish their normal applicator treatment lengths in order to absolve themselves of liability for an applicator not being that exact length (within a few millimeters). However, this policy backfires when applicator lengths are mismeasured by 10 cm and what could have been a small positioning error (~1-3 mm) becomes a large one with deleterious effects to the patient. Additionally, not enough patients are being imaged prior to the administration of their brachytherapy. Even if the institution is not doing 3-dimensional treatment planning, a fluoroscopic image should be taken with the catheter and dummy marker(s) in place to assure correct positioning of the applicator within the patient.

Many of the errors had very little to do with modern technology. In RP therapies, administering the dose to the correct patient was problematic. Right or left sidedness was an issue in both RP administration (left lobe versus right lobe) and in GK radiosurgery (wrong hemisphere). No events were reported that were specifically caused by modern computerized imaging techniques. There were no reports of the wrong patient's data being transferred and used for treatment planning, images being fused to other data sets incorrectly, or incorrect definition of applicators on computed tomographic scans. There was one instance of incorrect target definition on MRI. The use of modern equipment did cause errors, especially human errors and errors caused by lack of training with the use of the equipment. This was specifically apparent in the use of Y-90 microspheres and with new partial breast irradiation devices for HDRBT.

Almost all of the errors reported with prostate LDRBT could be attributed to lack of experience. While it dominated the error reporting in this time period, there are still debates about what constitutes a reportable medical event. The American Society for Radiation Oncology recently published guidelines that the definition of a medical event for prostate brachytherapy should be air-kerma strength related and not be dose based.[6] The NRC is currently gathering information through teleconferences and workshops to consider revising this definition. If and when the definition of a medical event is modified, it is possible that many of the events described here would no longer be considered a reportable event. However, regardless of the definition employed, it is clear by the D90 results in Fig 1 that there is room for improvement. Recent publications from the American Association of Physicists in Medicine include a report on quality assurance for ultrasound-guided implants[7] and recommendations on dose prescription and reporting for prostate implants[8] that may help lend guidance for improvements in LDRBT programs.

Conclusions

One hundred and forty-seven reported events were tallied and analyzed. Many of the events, such as the treatment of the wrong patient, did not have to do with modern technology. Errors that are historically frequent, like incorrect applicator length for HDRBT, are still occurring on a regular basis. Errors that did involve the use of modern technology were also relatively new methods of patient treatment, such as LDRBT and Y-90 treatment of the liver. Special care and training should be used when implementing these new procedures.

References

1. Bogdanich W. At VA Hospital, a Rogue Cancer Unit. The New York Times. Newspaper on-line. 2009. Available at: http://www.nytimes.com. Accessed November 15, 2010.
2. Method for evaluating QA needs in radiation therapy, American Association of Physicists in Medicine Task Group Report 100. 2009. (unpublished).
3. Ostrom LT, Leahy TJ, Novack SD, Summary of 1991 and 1992 misadministration events investigations. EG&G-2707, NUREG/CR-6088; 1994.

4. Thomadsen BR, Lin SW, Laemmrich P, et al. Analysis of treatment delivery errors in brachytherapy using formal risk analysis techniques. *Int J Radiat Oncol Biol Phys.* 2003;57:1492-1508.
5. Nuclear Regulatory Commission Event Notification Reports. Available at: http://nrc.gov/reading-rm/doccollections/event-status/event/. Accessed October 15, 2010.
6. Nag S, Demanes DJ, Hagan M, et al. Definition of medical event is to be based on the total source strength for evaluation of permanent prostate brachytherapy: A report from the American Society for Radiation Oncology [e-pub ahead of print]. *Practical Radiat Oncol.* doi: 10.1016/j.prro.2011.05.001. Accessed August 29, 2011.
7. Molloy JA, Chan G, Markovic A, et al. Quality assurance of U.S.-guided external beam radiotherapy for prostate cancer: Report of AAPM Task Group 154. *Med Phys.* 2011;38:857-871.
8. Nath R, Bice Jr WS, Butler WM, et al. AAPM recommendations on dose prescription and reporting methods for permanent interstitial brachytherapy for prostate cancer: Report of Task Group 137. *Med Phys.* 2009;36:5310-5322.